Interactive Learning tool for Dyslexia Kids

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Project Proposal Report

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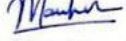
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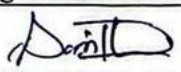
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Dyslexia is a neurodevelopmental learning disorder that mainly affects reading, spelling, language processing and working memory abilities. It is estimated that around 10 - 15% of school-aged children worldwide experience dyslexia and in Sri Lanka nearly 300,000 - 400,000 children show varying levels of dyslexic characteristics. One of the most challenging cognitive difficulties faced by dyslexic children is weak working memory. Working memory plays an important role in learning because it helps children remember sequences, follow instructions and process information while performing tasks. When this ability is weak, children often struggle to remember letters, numbers or instructions in the correct order, which significantly affects their learning progress in early education.

Although dyslexia is widely studied internationally, Sri Lanka currently lacks culturally relevant, technology-based learning tools specifically designed to improve working memory and sequencing skills for young dyslexic learners, particularly those in Grades 1-3. Most existing tools are designed for foreign educational contexts and are not aligned with the Sri Lankan primary school curriculum or bilingual learning environment.

To address this gap, this research proposes the Memory and Sequencing Module, which is developed as an individual component of a web-based gamified learning platform designed for dyslexic children aged 5-8 in Sri Lanka. The module focuses on improving working memory through a set of interactive digital games that are aligned with Grade 1-3 educational content, such as animals, fruits, alphabet letters, days of the week, and months. The activities are presented in both Sinhala and English to support bilingual learning. The games include digit-span exercises, sequence recognition tasks such as “What’s Missing?” and multi-step instruction-following activities. In addition, an adaptive difficulty mechanism dynamically adjusts the memory challenge based on the learner’s performance level.

To support learning improvement, the system incorporates visual and audio feedback mechanisms that guide the learner when mistakes occur and gradually reduce assistance as the learner becomes more accurate. A progress tracking system records performance indicators such as accuracy rates, error patterns and improvement trends, enabling a mastery-based learning progression.

The module will be implemented using React.js for the front-end interface, Node.js and Express.js for the back-end services and MongoDB for data storage, while Phaser.js will be used to develop the interactive game environment.

The expected outcome of this research is an engaging and effective digital intervention that enhances working memory, sequencing ability and instruction-following skills in young dyslexic learners, thereby supporting improved academic readiness and classroom participation. Beyond educational benefits, the system has potential social impact by supporting inclusive education for children with learning difficulties and commercial potential as a scalable educational technology solution for schools, learning centers and educational organizations in Sri Lanka.

Keywords: Dyslexia, Working Memory, Gamified Learning, Adaptive Learning, Sri Lanka

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I am also grateful to the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for providing the necessary academic environment, research resources and institutional support that made this project possible.

I would further like to extend my appreciation to external supervisor Dr. Kamalani Wanigasinghe for generously sharing her professional expertise on working memory difficulties among dyslexic children in Sri Lankan primary schools. Her professional insights played an important role in guiding and strengthening the development of the Memory and Sequencing Module.

Finally, this work is dedicated to children with dyslexia in Sri Lanka who face challenges related to memory and sequencing during learning activities. This research aims to contribute toward creating more supportive and accessible learning experiences for them through the proposed module.

TABLE OF CONTENT

Contents

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 7
LIST OF ABBREVIATIONS 8
1. INTRODUCTION 9
1.1 Background and Context 9
1.2 Literature Review 11
1.3 Research Gap 15
2. OBJECTIVE 16
2.1 Main Objective 16
2.2 Specific Objectives 16
3. METHODOLOGY 17
3.1 Research Approach 17
3.2 Requirement Analysis 17
3.3 Technical Methods and Algorithms 18
3.3.1 Adaptive Difficulty Engine 18
3.3.2 Scaffolded Error Correction 18
3.3.3 Mastery-Based Progression 18
3.3.4 Progress Tracking and Analytics 19
4. HIGH-LEVEL SYSTEM ARCHITECTURE 20
5. USER REQUIREMENTS 21
5.1 Stakeholder Identification 21
5.2 Functional Requirements 21
6. COMMERCIALIZATION PLAN 23
6.1 Target Market and Customer Persona 23
6.2 Value Proposition 23
6.3 Revenue Model 23
6.4 Competitive Advantage 23

6.5 Intellectual Property 23
7. BUDGET AND JUSTIFICATION 25
8. WORK BREAKDOWN STRUCTURE (WBS) 26
9. GANTT CHART 27
10. References 28
11. APPENDICES 30
Appendix A: Preliminary Survey Instrument for Educator Needs Analysis 30
Appendix B: Ethical Clearance Documentation 32
Appendix C: Preliminary UI Mockup Descriptions 33
Appendix D: Risk Analysis Register 34
Appendix E: Adaptive Algorithm Flowchart 35

LIST OF TABLES

Table 1 List of Abbreviations 8
Table 2 Literature Review Summary 14
Table 3 Budget and Justification ..... **Error! Bookmark not defined.**
Table 4 Work Breakdown Structure 26
Table 5 Gantt Chart 27

LIST OF ABBREVIATIONS

Table 1 List of Abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, 5th Edition
EEG	Electroencephalography
GBL	Game-Based Learning
IT	Information Technology
ML	Machine Learning
MSM	Memory and Sequencing Module
NLP	Natural Language Processing
RF	Random Forest
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SVM	Support Vector Machine
TAF	Topic Assessment Form
UI	User Interface
UX	User Experience
WM	Working Memory

1. INTRODUCTION

1.1 Background and Context

Dyslexia is one of the most common learning disabilities in the world, affecting somewhere between 5% and 15% of people globally [1]. For children with dyslexia, the struggle goes well beyond reading and writing. Many also find it genuinely difficult to process sounds, hold information in mind, follow sequences, and make sense of what they hear - challenges that touch nearly every part of their school day [1], [2]. In Sri Lanka, as in many developing countries, the support systems that these children need are hard to come by. Specialist services tend to be clustered in cities, diagnoses come late, and many families are simply left to manage on their own.

One of the most damaging - yet least talked about - aspects of dyslexia is what happens with working memory. Working memory is the mental workspace we all rely on to hold information while we are actively using it: following a teacher's instructions, keeping track of a number sequence, or connecting a new idea to something learned earlier. For children with dyslexia, this workspace is noticeably smaller than it is for their peers, and that gap quietly undermines their performance not just in literacy, but in maths and day-to-day learning too [1], [2].

What is striking, though, is how rarely this problem is directly addressed by the tools designed to help. Most digital dyslexia platforms focus on phonics, letter recognition, and reading practice - all worthwhile, but none of them systematically train working memory in an adaptive, meaningful way. Apps like Alexia Monsta [3], DyslexiaMate [4] and WORDEX [5] are genuinely promising steps forward, but none offer a dedicated working memory component that adjusts to each child's needs and connects to content they actually encounter in school. To make things harder, almost all of these platforms were built with English-speaking children in mind, making them feel foreign and irrelevant to Sri Lankan learners who study in Sinhala or Tamil and follow the national Grade 1-3 curriculum.

That gap is exactly what this research sets out to fill. The Memory and Sequencing Module (MSM) is a component of a wider dyslexia learning platform being developed collaboratively at SLIIT. Rather than retrofitting an existing tool, the MSM is built from the ground up to strengthen working memory through gamified activities that adapt to how each child is performing. All the content - the animals, fruits, months, days of the week, and alphabet sequences used in the games - comes directly from the Sri Lankan Grade 1-3 curriculum, so children are practicing real, familiar material while their memory skills are being quietly built up. The module also includes an error-correction system that does not just mark a wrong answer and move on but guides the child through a structured path toward getting it right [6], and a progression model that only advances the learner once genuine mastery has been demonstrated [7].

Ultimately, this work is about giving dyslexic children in Sri Lanka a fair chance at learning. It speaks directly to the United Nations Sustainable Development Goal 4, which calls for inclusive and equitable quality education for all. Building a tool that is affordable, locally

relevant and grounded in evidence is one concrete way to make that goal real for a group of children who have for too long, been left behind.

1.2 Literature Review

A review of existing studies on dyslexia interventions, working memory deficits, and assistive digital technologies shows that although there has been encouraging progress in supporting dyslexic learners, several important gaps remain. These gaps highlight the need for solutions that directly target the cognitive difficulties experienced by dyslexic children, particularly working memory limitations.

1.2.1 Dyslexia and Its Cognitive Profile

Dyslexia is a neurodevelopmental disorder that causes unexpected difficulties in reading, writing, and spelling, even in children who have normal intelligence and access to proper education [8]. However, dyslexia is not limited only to reading difficulties. Research shows that it affects several cognitive abilities simultaneously including phonological processing, working memory, auditory perception, visual discrimination, and executive functioning [1].

Patnoorkar et al. explain that dyslexia can appear in several forms such as primary, secondary, visual, auditory, and trauma-induced dyslexia. Their research also highlights that many dyslexic children show weaker short-term memory and working memory compared to their non-dyslexic peers [2]. This weakness in memory is not simply a side effect of dyslexia; instead, it plays a direct role in the reading and spelling challenges these children experience.

The neurological aspects of dyslexia have also been explored in recent research. Cura et al. studied changes in brain network activity among fourteen dyslexic children who participated in a 60-day cognitive training program. Using EEG-based functional connectivity and graph theory analysis, the researchers observed measurable improvements in both global and local brain network efficiency after the training [9]. These findings provide neurological evidence that structured cognitive training can positively influence brain organization in dyslexic children. This supports the idea that training working memory is not only educationally useful but also neurologically meaningful.

1.2.2 Assistive Technology and Digital Interventions for Dyslexia

Over the past decade, many technology-based tools have been developed to support learners with dyslexia. Patnoorkar et al. categorise these tools into four main groups: text-to-speech systems, eye-tracking technologies, virtual learning environments, and game-based applications [2].

Among these, text-to-speech tools such as *Read&Write* and the *IDEAL eBook reader* are widely used because they help students improve reading fluency and pronunciation. However, these tools mainly assist with the visible symptoms of dyslexia rather than addressing the underlying cognitive difficulties, particularly working memory limitations that many dyslexic children face.

Eye-tracking technology has also gained attention, especially for detecting dyslexia. Pinjari et al. reviewed multiple machine learning and deep learning systems that analyse eye movement patterns, handwriting, and speech data to identify dyslexia. Their results showed classification accuracy ranging from 85% to over 99%, depending on the approach used [10]. While these

systems are highly effective for identifying dyslexia, they mainly focus on diagnosis rather than providing long-term cognitive support. Once a child is identified as dyslexic, additional interventions are still required to help improve their learning abilities.

1.2.3 Game-Based and Gamified Learning for Dyslexia

Gamified learning approaches have shown strong potential in engaging dyslexic learners. Traditional teaching methods often struggle to maintain attention and motivation among children with learning difficulties, whereas game-based environments can make learning more interactive and enjoyable.

Gooch et al. examined how specialist dyslexia teachers used the *classDojo* platform with primary school students transitioning to secondary education [8]. Their findings showed that when rewards and badges were personalized according to each child's challenges, student motivation increased significantly. Students also became more aware of their own learning progress and developed a greater sense of ownership over their learning activities. The study highlights that the effectiveness of gamification depends not only on the platform itself but also on how educators use it.

Another example is *Alexia Monsta*, a 2D mobile application designed for dyslexic children aged four to eight. Developed using the ADDIE instructional design model, the application includes flashcard-based learning activities covering colors, shapes, and common household objects, supported by both visual and audio feedback [6]. Initial evaluation results showed positive learning engagement among children. However, the application does not adapt its difficulty level according to the learner's progress, nor does it specifically target working memory training. Additionally, its learning content is not aligned with any national school curriculum.

Ganegoda et al. developed *WORDEX*, a gamified learning system specifically designed for Sri Lankan primary school children aged seven to ten [5]. WORDEX uses Random Forest classifiers to identify four dyslexia subtypes and achieves an average classification accuracy of 97.62%. The platform includes gamified activities such as jumbled word games, spelling exercises, and sentence structuring tasks. While this system represents a significant advancement for Sri Lankan educational technology, its primary focus remains on reading-related challenges. Dedicated working memory training or sequencing support is not included in the platform.

1.2.4 Machine Learning Approaches to Dyslexia Assessment

Several researchers have also explored the use of machine learning techniques to improve dyslexia detection systems. Senthilselvi et al. developed a quiz-based classification model using Random Forest with Grid Search to assess dyslexia severity across five cognitive dimensions: language ability, memory, processing speed, visual discrimination, and auditory discrimination [6]. The system achieved a relatively low error rate of 5.8%. Although memory is included as one of the assessment factors, the platform does not provide any training activities aimed at improving working memory.

Similarly, Jency et al. proposed a two-stage quiz-based dyslexia prediction system that compares several machine learning algorithms including Decision Trees, Random Forest, Support Vector Machines (SVM), Logistic Regression, and AdaBoost [7]. Their results indicate that Random Forest with Grid Search and SVM with Grid Search achieved the highest performance. The system also includes parental guidance features and real-time feedback mechanisms, demonstrating a strong user-centered design. However, like many detection systems, its primary goal is identification rather than cognitive intervention.

Rajan et al. introduced another innovative approach by developing a hybrid AI-enabled SVM system implemented on a Raspberry Pi device. This system analyses children's speech signals to detect dyslexia and achieved a precision rate of 85% [11]. While this approach is technically innovative, it also focuses on diagnosis rather than providing continuous learning support after dyslexia is detected.

1.2.5 Multisensory and Adaptive Learning Platforms

Recent research has also explored multisensory learning platforms designed to support dyslexic learners. Faiza et al. introduced *DyslexiaMate*, a comprehensive system that combines air-writing recognition, an emotion-aware reading assistant, gamified learning activities, and a teacher monitoring dashboard [4]. The platform integrates technologies such as computer vision, natural language processing, and deep learning. Its air-writing recognition module achieved 93% accuracy, while the emotion-aware reading assistant reached 91% accuracy. The system's multisensory design, which integrates visual, auditory, and kinesthetic learning elements, resulted in a 25% increase in task completion rates compared to traditional learning methods. Despite these advanced features, DyslexiaMate does not include dedicated training activities specifically designed to improve working memory or sequencing skills.

Hany et al. proposed another system called *Nexia Tutor*, which combines a Random Forest screening service with a BERT-based tutoring system that delivers personalized, game-based language learning activities [12]. The screening system achieves an accuracy of 89.2%, while the tutoring system adapts content according to the learner's dyslexia subtype. Although the platform demonstrates improvements in phonological awareness and reading comprehension, working memory training remains outside the scope of its design.

Finally, Puritat et al. developed a language-independent gamified screening tool that uses auditory and visual matching tasks to distinguish between dyslexic and non-dyslexic children [1]. Their evaluation framework measures several performance indicators including response time, number of clicks, accuracy, and task efficiency. These metrics provide a useful model for evaluating cognitive performance and are particularly relevant for designing progress tracking systems in future interventions.

Table 2 Literature Review Summary

Year	Author	Product / System	Key Contribution	Gap (Working Memory Support)
2025	Faiza et al.	DyslexiaMate	AI-powered multisensory learning platform with air-writing recognition, reading assistant, and gamified activities.	Improves engagement but no dedicated Working Memory Support module.
2025	Ganegoda et al.	WORDEX	Machine-learning-based dyslexia screening and intervention platform for Sri Lankan primary students.	Focuses on reading and spelling; no working memory or sequencing training.
2025	Pinjari et al.	ML-based dyslexia detection	Uses eye-tracking, handwriting, and speech analysis with high accuracy.	Mainly diagnostic systems, not cognitive training platforms.
2024	Salim et al.	Alexia Monsta	Gamified mobile learning application with flashcards and multisensory learning.	No adaptive difficulty or working memory training module.
2024	Senthilselvi et al.	ML quiz-based dyslexia prediction model	Random Forest model predicts dyslexia severity including memory dimension.	Memory used for prediction only, not for training.
2024	Gooch et al.	ClassDojo gamification platform	Gamification increased motivation and engagement among dyslexic learners.	Focus on motivation; no structured working memory intervention.
2023	Patnoorkar et al.	Read&Write, IDEAL eBook Reader	Text-to-speech assistive tools widely used for reading support.	Support reading fluency only; no working memory support.
2023	Puritat et al.	Gamified dyslexia screening tool	Uses auditory and visual tasks to identify dyslexia.	Designed for screening, not working memory improvement.
2022	Various studies	Digital dyslexia learning tools	Introduced early gamified and assistive learning platforms.	Mostly focused on reading and phonological training.
2021	Various studies	AI-based dyslexia detection research	Early research applying ML models to detect dyslexia patterns.	Primarily diagnostic research, not intervention systems.

1.3 Research Gap

Looking across the research reviewed, several clear gaps become evident. Together, these gaps highlight the need for a solution like the Memory and Sequencing Module (MSM).

One major gap is the lack of direct support for working memory. Platforms such as Alexia Monsta [3], DyslexiaMate [4], WORDEX [5], and Nexia Tutor [12] have made valuable contributions to supporting dyslexic learners. However, their main focus is on reading skills, phonological awareness, and letter recognition. None of these systems provide a dedicated, gamified and adaptive module specifically designed to train working memory and sequencing skills. This is an important limitation because working memory difficulties often create some of the most common day-to-day challenges for dyslexic children, particularly when they are trying to follow multi-step instructions or keep up with classroom activities.

Another important gap is the lack of locally relevant learning content. Many of the international platforms reviewed were developed without considering any specific national curriculum. As a result, a child in Sri Lanka using platforms such as DyslexiaMate [4] or Nexia Tutor [12] would often encounter learning materials that feel unfamiliar or disconnected from what they are studying in school. Currently, there is no system that combines working memory training with Sri Lankan Grade 1- 3 curriculum content, meaning children lose the opportunity to strengthen their cognitive skills while also reinforcing what they learn in their classrooms.

A further limitation lies in how difficulty levels are adjusted in existing platforms. Most systems either keep difficulty fixed or adjust it only occasionally based on general performance categories. In reality, learning ability can change even within a single session and support should adapt accordingly. The MSM addresses this by adjusting the memory load dynamically, increasing the sequence length step by step based on the child's performance. For example, tasks gradually move from two items to three and then to four items only when the child demonstrates consistent improvement.

Another issue relates to the type of feedback provided to learners. In many existing systems, when a child answers incorrectly, the platform simply marks the response as wrong and moves on. This approach does not help the child understand what went wrong or how to improve. In contrast, the MSM provides guided corrective feedback. When an error occurs, the system identifies where the mistake happened, demonstrates the correct answer using clear visual and audio cues, and then generates targeted practice activities to help the child strengthen that specific skill. Over time, this support gradually decreases as the learner gains confidence and accuracy.

Finally, progression in many platforms is often based simply on task completion rather than true mastery. Learners may move to higher levels even if their understanding is incomplete. This can create weak learning foundations. The MSM takes a different approach by allowing children to progress only after they have shown consistent accuracy across multiple attempts, ensuring that each level is genuinely understood before moving forward. This approach encourages meaningful learning and builds stronger confidence in the learner's abilities.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate an adaptive, gamified Memory and Sequencing Module that systematically strengthens working memory and instruction-following skills in dyslexic children aged 5-8 through curriculum-aligned, mastery-based interactive activities.

2.2 Specific Objectives

1. To develop a suite of memory-based gamified activities - including digit-span tasks, 'What's Missing?' exercises, and multi-step instruction-following games - using Sri Lankan Grade 1-3 curriculum content (animals, fruits, the alphabet, days of the week, and months of the year).
2. To implement an adaptive difficulty engine that dynamically adjusts the working memory load (from 2 to 3 to 4 or more items) based on real-time analysis of a learner's accuracy and consistency metrics.
3. To design and integrate a scaffolded error-correction mechanism that identifies specific failure points in a learner's performance, provides reinforced visual and audio feedback with correct answers and generates targeted follow-up practice until consistent mastery is achieved.
4. To develop a mastery-based progression system that restricts advancement to higher difficulty levels until a child demonstrates sustained, consistent accuracy across a statistically significant number of trials.
5. To build a comprehensive progress-tracking subsystem that records accuracy rates, error patterns, attempt counts, and skill improvement trajectories for use by parents, teachers, and the adaptive engine.
6. To validate the module's effectiveness in improving working memory capacity and instruction-following ability in dyslexic Sri Lankan children through a rigorous mixed-methods evaluation study.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach, which is well suited for a project that focuses on designing and evaluating a practical solution. DSR treats the process of building a system and evaluating its effectiveness as interconnected activities rather than separate steps. In this study, the Memory and Sequencing Module (MSM) will be developed while simultaneously being evaluated for its effectiveness in supporting working memory training among dyslexic children.

The research will progress through five main phases. It will begin by understanding the problem and reviewing existing solutions. This will be followed by system design and early prototyping. The next phase involves developing and integrating the full system, after which the module will be tested and refined through user interaction and feedback. Finally, the study will conclude with a comprehensive evaluation and documentation of the findings.

These phases are not treated as separate stages. Feedback obtained at any point during development may lead to improvements or adjustments in earlier stages. This iterative process ensures that the final system reflects the educational, cognitive, and cultural context of Sri Lankan dyslexic learners, rather than simply applying a generic technological solution.

3.2 Requirement Analysis

Before the system design begins, the research will focus on carefully identifying the requirements of the proposed solution. This stage involves collecting insights from multiple stakeholders who understand the challenges faced by dyslexic children.

Structured interviews and focus group discussions will be conducted with three key groups: specialist dyslexia teachers and educational psychologists working within the Sri Lankan education system, parents of children in Grades 1-3 who show signs of dyslexia or have been diagnosed with it, and the children themselves. Participation from children will take place only with full parental consent, as they are the primary users of the module.

In addition to stakeholder discussions, official Grade 1-3 curriculum documents from the Sri Lanka National Institute of Education (NIE) will be reviewed. This ensures that the activities designed for the MSM are aligned with the learning material that children are already exposed to in school. By linking cognitive training with familiar curriculum content, the system aims to support both memory development and classroom learning simultaneously.

3.3 Technical Methods and Algorithms

3.3.1 Adaptive Difficulty Engine

A key feature of the MSM is its adaptive difficulty engine, which personalizes the learning experience for each child. The system continuously monitors the child's performance and adjusts the difficulty level accordingly.

Instead of making changes based on a single attempt, the algorithm evaluates the child's rolling accuracy across their five most recent attempts. If a child achieves more than 80% accuracy across two consecutive blocks, the system increases the memory load by one item, gradually increasing the challenge. Conversely, if accuracy falls below 50%, the memory load is reduced and the system activates a remediation sequence to provide additional support.

This responsive, threshold-based mechanism allows the difficulty level to change dynamically based on real performance. The approach is inspired by adaptive learning models discussed by Senthilselvi et al. [6] and Jency et al. [7], which demonstrate the effectiveness of fine-grained adaptive systems in supporting dyslexic learners.

3.3.2 Scaffolded Error Correction

In the MSM, making a mistake is treated as a learning opportunity rather than a failure. When a child answers incorrectly, the system activates a three-step remediation process designed to guide the child toward the correct understanding.

First, the system clearly presents the correct answer using both visual animation and audio explanation, helping the child understand the correct response. Second, the same question is presented again, but this time with a visual cue highlighting the correct answer to support the learner. Third, these cues are gradually removed in subsequent trials until the child can answer correctly without assistance.

This process follows the educational principle known as Vygotsky's Zone of Proximal Development, which suggests that children learn most effectively when they receive guided support that gradually decreases as their skills improve.

3.3.3 Mastery-Based Progression

The MSM also uses a mastery-based progression model. In this system, completing an activity alone is not enough to move to the next level. Instead, the child must demonstrate consistent understanding of the skill.

To unlock the next difficulty level, the child must achieve at least 80% accuracy across two independent trial blocks at the current level. This requirement ensures that children develop a solid understanding before moving forward. Without such safeguards, learners may progress too quickly and struggle later because the foundational skills were not fully developed.

The 80% mastery threshold is widely used in adaptive learning research and supports the core philosophy of the MSM: meaningful learning progress should be based on demonstrated understanding rather than simple task completion.

3.3.4 Progress Tracking and Analytics

Every interaction within the MSM generates useful data about the child's learning progress. The system uses Firebase Firestore to store trial-level information such as accuracy scores, error types, response time, and the number of attempts required to complete a task.

This data is visualized through a dashboard designed for teachers and parents. The dashboard presents information through accuracy trend charts, error pattern heatmaps, and level progression timelines. These visual summaries help educators and parents easily understand how the child is progressing and identify areas that may require additional support.

The design of this tracking system is inspired by the performance measurement model used by Puritat et al. [1], which recorded metrics such as total clicks, task duration, accuracy, and efficiency in a gamified dyslexia assessment system. Their work demonstrates how performance data can be interpreted effectively to support informed intervention strategies.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

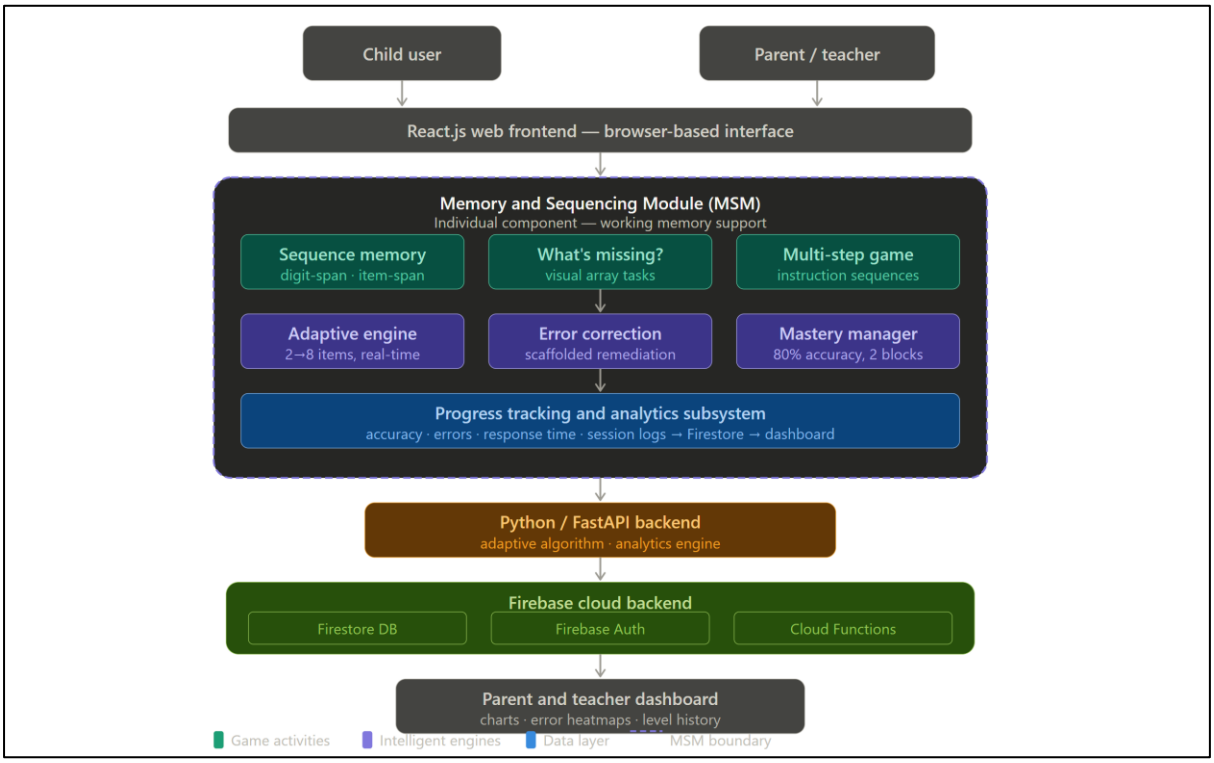


Figure 1 Overall component Architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Primary Users: Dyslexic children aged 5-8 enrolled in Sri Lankan Grades 1-3.

Secondary Users: Parents and legal guardians who configure child profiles and monitor progress.

Tertiary Users: Class teachers and specialist dyslexia educators who review analytics and adapt instructional strategies.

System Administrators: SLIIT research team responsible for platform maintenance and curriculum content updates.

5.2 Functional Requirements

1. The system shall provide at least three distinct gamified activity types targeting working memory: sequence recall, item detection, and multi-step instruction following.
2. The system shall use Sri Lankan Grade 1-3 curriculum content (animals, fruits, the alphabet, days of the week, months of the year) as the semantic basis for all game stimuli.
3. The system shall implement an adaptive difficulty engine that adjusts memory load (minimum 2 items, maximum 8 items) based on rolling accuracy metrics computed over five consecutive trials.
4. The system shall present the correct answer with visual and audio reinforcement whenever a child responds incorrectly.
5. The system shall generate a follow-up practice sequence of structurally identical items following each error, progressively reducing cueing support until the child answers correctly without assistance.
6. The system shall prevent level advancement until the learner achieves at least 80% accuracy across two independent consecutive trial blocks.
7. The system shall record per-trial accuracy, error type, response latency, and attempt count, storing all data securely in the cloud database.
8. The system shall provide a parent and teacher dashboard displaying accuracy trends, error pattern heatmaps, and progression histories.
9. The system shall support bilingual audio narration in Sinhala and English.
10. The system shall operate in offline mode, catching session data locally and synchronizing upon connectivity restoration.

5.3 Non-Functional Requirements

Usability: The interface shall be navigable by a child aged 6 with no prior experience, using icon-based navigation with audio instructions and no reliance on text reading.

Performance: Game activities shall launch within 2 seconds; adaptive difficulty recalibration shall complete within 500 milliseconds of block completion.

Reliability: The system shall maintain a crash-free session rate of at least 99% in production.

Security: All child data shall be pseudonymised, stored with field-level encryption, and accessible only to authorised parent and teacher accounts.

Accessibility: All visual game elements shall comply with WCAG 2.1 contrast ratio guidelines; all instructions shall be available in audio format.

Scalability: The backend architecture shall support concurrent use by at least 500 active learner sessions without performance degradation.

Maintainability: Curriculum content (stimuli lists) shall be managed via a CMS-style admin interface to allow non-technical educators to update content without code changes.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The main target market for the MSM is parents of dyslexic children aged 5-8 in Sri Lanka who are looking for affordable and effective digital learning support for their children.

The system can also be useful for several secondary users, including:

- Government and private primary schools that want inclusive learning tools for students with learning difficulties
- Specialist dyslexia learning centres that provide additional support for children with dyslexia
- The Sri Lanka Ministry of Education, which may adopt the system in the future as part of national inclusive education initiatives.

6.2 Value Proposition

The MSM provides a unique learning solution for Sri Lankan dyslexic children.

It is the first gamified and adaptive working memory training tool designed specifically for children following the Sri Lankan Grade 1–3 curriculum.

The platform is available in both Sinhala and English, making it accessible to more students.

Unlike many international learning applications, the MSM includes:

- Culturally familiar learning content
- Activities based on the Sri Lankan school curriculum
- Mastery-based progression, where children move forward only after understanding the concept
- Step-by-step error correction support that helps children learn from mistakes.

6.3 Revenue Model

School Subscription: Schools pay a yearly fee per student to access the full module and teacher monitoring features.

Freemium for Parents: Parents can download a free basic version, while the full version is available through a low-cost monthly subscription.

6.4 Competitive Advantage

Unlike international competitors such as Nessy Learning or Alexia Monsta, the MSM is purpose-built for the Sri Lankan educational context, curriculum-aligned, and freely accessible on low-cost Android devices. Its adaptive working memory focus addresses a gap untouched by existing local platforms including WORDEX and Arunalu.

6.5 Intellectual Property

The MSM platform will follow the intellectual property policies of SLIIT.

The following components will be protected:

- Source code of the system
- Adaptive learning algorithms
- Curriculum-based learning content

Some research-related components may be released as open-source to support academic collaboration. However, the complete platform and commercial deployment rights will remain protected.

7. BUDGET AND JUSTIFICATION

Table 3 Budget And Justification

Budget Item	Estimated Cost (LKR)	Justification
Android Test Device (budget tablet)	Available	Required for on-device UX testing with child participants
Firebase Blaze Plan	5,000	Supports Firestore, Cloud Functions, and Authentication at scale
Sinhala Audio Recording (professional)	20,000	Professional voice talent for bilingual audio narration of all game content
Google Cloud TTS API	5,000	For Sinhala and English audio generation
UI Asset Design (icons, animations -Figma)	6,000	Custom child-friendly game assets; cannot use stock assets for curriculum-specific items
Participant Incentives (stationery)	5,000	Ethical incentives for child participants and guardians during evaluation sessions
Printing and Documentation	1,000	Consent forms, assessment instruments, and reporting materials
Research and Travel cost	15,000	School visits for user testing and data collection
TOTAL	57,000	

8. WORK BREAKDOWN STRUCTURE (WBS)

The following table outlines the task breakdown for the Memory and Sequencing Module across the project timeline.

Table 4 Work Breakdown Structure

	Task	**Deliverable**	**Duration**
1	Literature Review and Needs Analysis	Annotated bibliography; needs analysis report	3 weeks
2	System Requirements Specification	SRS document for WM module	2 weeks
3	Game Design and UI Wireframing	Wireframes and game design document	2 weeks
4	Front-End Development (React + Phaser.js)	Functional game UI prototypes	5 weeks
5	Back-End API Development (Node.js + Express)	RESTful API endpoints for game logic	4 weeks
6	Database Design and Integration (MongoDB)	Configured database schema and connection	2 weeks
7	Adaptive Algorithm Implementation	Working adaptive difficulty engine	3 weeks
8	Scaffolded Error Correction System	Implemented feedback and retry mechanism	2 weeks
9	Progress Tracking and Dashboard	Educator dashboard with analytics visualization	3 weeks
10	Bilingual Content Integration	All Sinhala and English audio and text assets	2 weeks
11	Unit and Integration Testing	Test reports; bug-fix log	3 weeks
12	User Testing and Evaluation	Usability test report; efficacy evaluation report	3 weeks
13	Final Report and Documentation	Complete proposal and technical report	2 weeks

9. GANTT CHART

Table 5 Gantt Chart

Task	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
1. Literature Review & Needs Analysis	■	■								
2. Requirements Specification		■	■							
3. Game Design & Wireframing			■	■						
4. Front-End Development				■	■	■				
5. Back-End API Development				■	■	■				
6. Database Integration					■	■				
7. Adaptive Algorithm					■	■	■			
8. Error Correction System						■	■			
9. Progress Tracking & Dashboard							■	■		
10. Bilingual Content Integration							■	■		
11. Testing								■	■	
12. User Evaluation									■	■
13. Documentation & Final Report									■	■

Legend: Shaded cells (■) indicate active project phases for the corresponding month.

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11. APPENDICES

Appendix A: Preliminary Survey Instrument for Educator Needs Analysis

The following survey was developed for administration to primary school teachers in the Western Province of Sri Lanka. Its purpose is to understand current classroom practices, identify the specific challenges dyslexic learners face, and gather teacher perspectives on what a digital working memory support tool would need to offer to be genuinely useful in the local context.

Section 1 - Classroom Context

Question 1: How many students in your current class have been formally or informally identified as having reading difficulties or dyslexia? Please provide an approximate number.

Question 2: On a scale of 1 to 5, how confident do you feel in identifying working memory difficulties in your students, separate from general reading or attention problems?

Question 3: How often do your struggling readers have difficulty following multi-step verbal instructions in class?

Never / Rarely / Sometimes / Often / Almost always

Section 2 — Working Memory and Sequencing Challenges

Question 4: Which of the following do you observe most frequently in students who struggle with reading or instruction-following? (Select all that apply)

Forgetting the beginning of an instruction by the time it is finished

Difficulty recalling sequences of letters, numbers, or steps

Losing their place during reading or copying tasks

Struggling to remember what was taught in the previous lesson

Confusion when asked to follow more than two instructions at once

Question 5: What specific classroom activities do your dyslexic or struggling learners find most difficult when it comes to remembering and sequencing information? Please describe briefly.

Question 6: What strategies or tools do you currently use to support working memory in these students? How effective have they been?

Section 3 - Digital Tools and Technology Readiness

Question 7: How familiar are you with digital learning applications designed specifically for dyslexic learners?

Not familiar at all / Slightly familiar / Moderately familiar / Very familiar

Question 8: Have you used any digital tools to support literacy or memory skills in your classroom? If yes, please name the tool and describe your experience briefly.

Question 9: What concerns, if any, do you have about introducing a tablet or computer-based learning activity into your classroom for students with dyslexia?

Section 4 - Cultural and Linguistic Considerations

Question 10: In your view, what cultural or linguistic features would make a digital learning tool most effective for Sri Lankan children in Grades 1–3? Please consider language of instruction, familiar content themes, and visual design.

Question 11: How important is it that the content of a learning application reflects the Sri Lankan national curriculum (e.g., using familiar animals, fruits, months, and alphabet content from Grade 1–3 textbooks)?

Not important / Slightly important / Important / Very important / Essential

Question 12: Is there anything else you would like to share about the needs of dyslexic learners in your classroom that you feel is not captured above?

Appendix B: Ethical Clearance Documentation

An application for ethical clearance has been submitted to the SLIIT Research Ethics Committee in accordance with the institution's guidelines for research involving human participants, with particular attention to the protections required for child participants under the age of 18.

The application package includes the following components:

(a) Participant Selection and Recruitment Procedures - A description of the inclusion criteria for child participants (aged 6–10, enrolled in Sri Lankan Grades 1–3, with a formal or provisional dyslexia diagnosis from a qualified educational psychologist or paediatrician), the recruitment channels to be used (partner primary schools and specialist learning centres in the Western Province), and the procedures for ensuring voluntary participation without coercion.

(b) Informed Consent and Assent Documentation - Parental and guardian informed consent form templates prepared in both Sinhala and English, clearly explaining the purpose of the study, what participation involves, how data will be stored and used, the child's right to withdraw at any time without consequence, and the contact details of the supervising researcher and ethics committee. A child-friendly verbal assent script is also included for use during face-to-face sessions.

(c) Data Management and Anonymization Plan - A description of how all participant data will be pseudonymized using UUID-based identifiers from the point of collection, how personally identifiable information will be stored separately from session data under restricted Firebase authentication rules, the data retention period (24 months post-study), and the procedures for secure deletion upon expiry.

(d) Risk Assessment and Mitigation Strategies - An identification of potential psychological risks (e.g., frustration or distress arising from repeated task failure) and academic risks (e.g., disruption to regular schooling), alongside the mitigation strategies in place, including session time limits, teacher oversight during sessions, and the researcher's protocol for pausing or ending a session if a child shows signs of distress.

Ethical clearance is currently pending. Approval is anticipated prior to the commencement of participant recruitment. Upon receipt, the signed clearance certificate will be appended to this section. Participant recruitment will not begin under any circumstances before formal clearance is granted.

Appendix C: Preliminary UI Mockup Descriptions

Preliminary wireframes for the Memory and Sequencing Module game interfaces have been developed in Figma following an initial round of stakeholder consultations with two primary school teachers and one specialist dyslexia educator. The mockups are guided by dyslexia-friendly interface design principles including high-contrast colour schemes, icon-based navigation, large touch targets, and minimal on-screen text to reduce reading demands.

The following screens have been wireframed and are available for review:

(a) Module Home Screen and Game Selection Menu - A welcoming landing screen displaying three large, icon-labelled game tiles representing the three activity types: Sequence Memory, What's Missing?, and Multi-Step Instructions. Each tile uses a distinct colour and a simple pictographic icon so that children who cannot yet read can navigate independently. A progress indicator showing unlocked versus locked levels is visible at the top of the screen.

(b) Sequence Memory Game Interface - Difficulty Levels 2, 3, and 4 - Three wireframe variants showing how the stimulus presentation area scales with memory load. At level 2, two large item cards are shown sequentially in the centre of the screen. At level 3, three cards appear. At level 4, four cards are shown. After the stimulus sequence disappears, response cards appear at the bottom of the screen for the child to tap in the correct order. A large audio replay button is positioned in the top-right corner.

(c) What's Missing? Game Interface - A grid of familiar curriculum objects (e.g., six animals from the Grade 1–3 syllabus) is displayed, with one item absent. The child selects the missing item from three options shown below the grid. Difficulty increases by expanding the grid size and reducing the visual distinctiveness of the distractor options.

(d) Multi-Step Instruction Game Interface - An animated character on screen delivers a verbal instruction sequence (e.g., "tap the fruit, then clap, then find the letter A"). The instruction is accompanied by a visual sequence strip at the top of the screen showing the steps as icons. The child completes each step by interacting with the on-screen objects.

(e) Error Correction Feedback Screen - When an incorrect response is given, a calm, non-punitive feedback animation plays. The correct answer is highlighted with a soft glow and the audio narration repeats it clearly. A "try again" button appears with the same question, this time with the correct answer position subtly cued through a coloured border. This screen was specifically designed to avoid any imagery or sounds that could feel discouraging or alarming to a young child.

All mockups will be iteratively refined based on feedback gathered during the formative usability testing phase with the target age group. Final high-fidelity designs will replace these wireframe descriptions in the complete dissertation submission.

Appendix D: Risk Analysis Register

The following register identifies the key risks associated with the development and evaluation of the Memory and Sequencing Module, along with their assessed likelihood and impact ratings and the specific mitigation strategies in place.

Risk	Likelihood	Impact	Mitigation
Difficulty recruiting sufficient child participants for user testing	Medium	High	Early engagement with three partner primary schools and two specialist learning centres in the Western Province; provision of bilingual (Sinhala/English) parental information packs; flexible session scheduling to accommodate school timetables
Adaptive algorithm not achieving desired personalisation performance	Low	High	Iterative algorithm refinement during development phase using simulated performance data before live testing; thresholds configurable via admin interface without code changes; consultation with adaptive learning literature to validate threshold values
Sinhala content accuracy errors in game stimuli or audio narration	Low	Medium	All Sinhala-language content reviewed and approved by a qualified Sinhala language educator prior to integration; audio narration recorded by a native speaker and verified by a second reviewer
Cloud infrastructure (Firebase) costs exceeding allocated budget	Low	Medium	Development conducted entirely within Firebase free tier; Blaze plan upgrade budgeted as a line item; all costs monitored monthly against budget allocation
Low learner engagement with module content during evaluation	Medium	High	Gamification design directly informed by child engagement research reviewed in the literature; iterative UI testing with target age group before full evaluation; teacher and parent involvement in session delivery to maintain motivation
Child participant withdrawal during the eight-week intervention	Medium	Medium	Sample size calculation accounts for 20% attrition; ongoing parent communication and relationship management; sessions kept brief (15–20 minutes) to minimise burden
Network connectivity issues affecting data synchronisation during sessions	Low	Medium	Offline-first architecture ensures all session data is cached locally and synchronised to Firebase upon connectivity restoration; no session data is lost due to connectivity loss

Appendix E: Adaptive Algorithm Flowchart

The following describes the complete logic of the adaptive difficulty algorithm that governs the Memory and Sequencing Module. All values stated here are consistent with those defined in Chapter 3 of this proposal. A diagrammatic flowchart version will be included as a Figma export in the final dissertation submission.

Inputs: Current difficulty level N (memory load, measured in number of items), rolling session accuracy data stored in Firebase Firestore, block completion events.

Algorithm Steps:

Step 1 - Load learner state Retrieve the child's current difficulty level (N items, where minimum N = 2 and maximum N = 8) and the accuracy records for the most recent trial blocks from Firebase Firestore.

Step 2 - Evaluate for difficulty increase If the child achieves accuracy ≥ 80% across two consecutive independent trial blocks at the current level N, increment N by 1 (e.g., from 3 items to 4 items), up to the defined maximum of N = 8. Reset the consecutive block counter.

Step 3 - Evaluate for difficulty decrease If the child achieves accuracy < 50% across two consecutive independent trial blocks at the current level N, decrement N by 1 (e.g., from 3 items to 2 items), down to the defined minimum of N = 2. Trigger the scaffolded error correction sequence. Reset the consecutive block counter.

Step 4 -Maintain current level If neither threshold in Steps 2 or 3 is met, maintain the current difficulty level N and present the next game instance at the same level without adjustment.

Step 5 - Handle individual trial errors On any single incorrect trial response, trigger the error correction mechanism immediately: (a) display and aurally announce the correct answer with animated visual reinforcement; (b) present a structurally identical question with partial cueing; (c) gradually withdraw cueing across subsequent attempts until the child answers correctly without support.

Step 6 - Update session record At the conclusion of each trial block, write a session record to Firebase Firestore containing: trial results, accuracy rate for the block, response latency values, updated difficulty level N, timestamp, and any error correction events triggered.

Step 7 - Evaluate mastery completion If the child achieves accuracy ≥ 80% across two consecutive trial blocks at the maximum level (N = 8), the Mastery and Progression Manager trigger a mastery completion event, generates a visual completion reward within the app, and unlocks the next available module within the platform.

Key algorithm parameters

Parameter	Value
Minimum memory load (N)	2 items
Maximum memory load (N)	8 items

Advancement threshold	≥ 80% accuracy
Consecutive blocks required to advance	2 blocks
Reduction threshold	< 50% accuracy
Consecutive blocks triggering reduction	2 blocks
Mastery criterion	≥ 80% across 2 blocks at N = 8
Database	Firebase Firestore